

Gaurav Raj

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PROFILE

Offensive security researcher and published author specialising in **red team operations, Active Directory / Kerberos, reverse engineering, and kernel development**. Author of the book *Forged Trust* and a full-taxonomy ADCS whitepaper, running penetration testing engagements end to end from client contact through to delivery. Creator of BerserkArch Linux, focused on secure, automated, and scalable offensive security environments. Supported by deep low-level systems experience (C/Assembly/Rust) and Linux internals.

PUBLICATIONS & ACADEMIC CITATIONS

Book: *Forged Trust - Offensive Operations Against ADCS*. Comprehensive coverage of ESC1-18, THEFT1-5, and (D)PERSIST.
Paper: *Certificate of Compromise*, a full-taxonomy ADCS SoK / whitepaper. DOI: 10.5281/zenodo.20732733.

TECHNICAL SKILLS

Red Team & Pentest: Reverse Engineering, Network/Internal/Web Pentesting, AD Attacks, C2 (Cobalt Strike/Sliver).
AI & LLM: LLM Training & Fine-tuning, Prompt Engineering, Model Architecture, Evaluation, LLM Red Teaming.
Tooling: NetExec, Impacket, bloodyAD, Sliver C2, BloodHound, Certipy, Ghidra, IDA Pro, BurpSuite Pro.
Core Languages: C, C++, Assembly, Rust, Golang, Python, Bash, Zig, JavaScript, TypeScript, Java.
Systems Engineering: Linux Internals, Kernel Dev, Memory Management, Bootloaders, Device Drivers, Embedded/IoT.
Cloud & DevOps: AWS, Azure, GCP, Terraform, Vagrant, Kubernetes, Docker, Github Actions, CI/CD, Linux.

SECURITY & ENGINEERING EXPERIENCE

BerserkArch Linux Distro Project <i>Lead Maintainer, Developer & Systems Architect</i>	Remote <i>2025 - Present</i>
OS: Designed and maintain an Arch Linux-based distribution for red team operations and secure development. CI/CD: Built fully automated ISO CI/CD pipelines using GitHub Actions and Docker (build, sign, checksum, deploy). Containerisation: Published official Docker images for portable offensive security labs and rapid development. Security Hardening: Implemented custom kernel patches and AppArmor profiles for system hardening.	

Cybercraft Labs <i>Independent Security Consultant & Engineer</i>	Remote <i>Sept 2023 – Feb 2026</i>
Penetration Testing: Delivered network, web, and cloud penetration testing engagements for clients end to end, running each from initial contact and scoping through to reporting and remediation guidance. Cloud Hardening: Hardened client AWS/GCP hybrid infrastructure, including EC2 hardening and SES email authentication (DKIM/DMARC). Software Development: Designed and delivered bespoke client software to specification, including full-stack web applications, desktop tools, and custom reconnaissance automation that streamlined engagement workflows.	

KEY R&D PROJECTS

Vader LLM <i>github.com/thehackersbrain/vader</i>	<i>Language Model Research & Engineering</i>
GPT-style language models implemented and trained from scratch, exploring dense architecture, and instruction tuning.	
MalDev Wiki <i>https://maldev.thehackersbrain.dev</i>	<i>Malware Development Research and Wiki</i>
Practical research and implementation of malware-development techniques, evasion and related offensive tradecraft.	
THBOS (Custom x86 Kernel) <i>Source Code</i>	<i>Systems Programming (C, Assembly & Zig)</i>
64-bit OS kernel from scratch with custom bootloader, Protected Mode, CPUID-driven topology.	
HiddenWave <i>github.com/thehackersbrain/hiddenwave</i>	<i>Steganography Tool (C++ & Python)</i>
LSB audio steganography engine with WAV/MP3 support. Featured in the Netflix series Gyeongseong Creature (S2E1).	

ACHIEVEMENTS

- VulnHub:** Designed 2 vulnerable VMs (*Blogger, CorpVision*) with custom exploit chains (Webmin, Cronjob privesc).
- Advanced Security Labs:** Completed **HackTheBox Fortresses** (Jet) & **TryHackMe Networks** (Wreath, Holo), mastering pivoting, AD exploitation, and perimeter breaching.
- Global Rankings:** **Top 1% on TryHackMe** (CTF) | **HackerRank Gold Badge** (Problem Solving & Algorithms).

EDUCATION

B.Sc (Maths Hons) <i>Specialisation:</i>	<i>Patliputra University, Patna</i> <i>Advanced Mathematics (i.e. Calculus, Linear Algebra)</i>	2027 (ongoing) -
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